

Joby Aviation: Master Interview Preparation Document

Candidate: Advait Chordia

Interviewer: David Pedraza

Role: Composite Manufacturing Internship

1. About Me

Tell me about yourself:

I'm a mechanical engineering student at the University of Illinois studying engineering mechanics with a minor in materials science, and most of my experience has been focused on composite structures and how they're actually manufactured.

I first became interested in composites growing up around cars and hearing about carbon fiber as this almost mysterious material. What really hooked me was realizing that the real challenge isn't just the material itself — it's figuring out how to design and manufacture parts so that the material actually performs the way it's supposed to.

At Illinois Electric Motorsport, I lead the structural development of our composite front wing and help coordinate manufacturing for a 21-element aero package. That's given me a lot of hands-on experience with prepreg layups, vacuum bagging, and writing SOPs so that a larger team can reliably manufacture parts. A big part of that role has been learning from issues we encountered last year — things like bridging during layup — and improving the manufacturing process so we can build more reliable parts this season.

Alongside that, I worked in a composites research lab where I characterized continuous-fiber composites using ASTM tensile and shear testing, and worked with continuous-fiber 3D printing systems to control fiber alignment and deposition. That experience helped me understand how processing decisions directly affect material behavior.

What excites me about Joby is that it's the perfect next step from the work I've been doing — moving from student-scale composite fabrication into a professional manufacturing environment where those same design, materials, and process challenges are being solved to build certified flight hardware.

- My experiences are focused towards composites. Composite part design for light, high load facing parts, characterizing materials and more experience with physical part manufacturing.
- I think what sets me apart as a composite manufacturing engineer would be my experience designing these parts and manufacturing them, realizing the possible gaps between the two.
- I've worked with a team to manufacture a 21 element aero package last year and we are in the process of manufacturing our parts this year. I've had the opportunity to use my learnings from mistakes made last year to remedy them and make much better, high functioning and dependable parts this year.
- I am interested in Joby specifically because I think it represents the perfect next step in my professional career, moving from the world of student level prototyping and research to a professional environment where I can help solve the real-world manufacturing challenges of bringing the first certified eVTOL to market.

~~"I am a structures and composites focused Mechanical Engineer who thrives at the intersection of high-fidelity simulation and hands-on manufacturing. What sets me apart is my 'builder's intuition'—the ability to identify precisely where a design will fail on the shop floor and proactively standardize processes to ensure quality and efficiency."~~

~~Connecting to Joby's Needs: "I understand that Joby is currently navigating the critical transition from low-volume prototyping to certified, high-rate production. As the Structures Lead for my FSAE team, I successfully led a 9-member crew to reduce carbon fiber waste by 70% through automated nesting and implemented Root Cause Analysis (RCA) to eliminate structural defects like bridging in complex split molds. These are exactly the types of process-driven efficiencies and quality control measures required to move an aircraft from a lab environment to a certified production line."~~

~~The Future: "I am interested in this role at Joby because it represents the perfect next step in my career: moving from the world of student prototyping and additive research into a professional environment where I can help solve the real-world manufacturing challenges of bringing the first certified eVTOL to market."~~

Why Joby:

- Joby is in a place where they're going from prototyping to actual scaled production. It's in a pivotal point and I think that right now would be an incredible time to join.
- It is in an industry that is trying to do something spectacular and is on the frontier of that. Considering my experiences in high paced environments with having to stick with tight timelines, as well as my experience in the industry (Force) I think I've found that this is the direction I want to propel my career in.

2. STAR Interview Stories

2.1 FSAE

2.1.1 SOP Layup Manuals & Material Optimization (FSAE)

Situation: Our 21-element aero package build relied heavily on "tribal knowledge" and a 9-member team with varying levels of experience. This lack of documentation led to high scrap rates and inconsistent part weights, which threatened our tight manufacturing window.

Task: My goal was to move from a "craft" mindset to a "production" mindset by standardizing our manufacturing process to ensure that Part #1 and Part #21 were identical in quality and weight.

Action: * **Documentation:** I developed a comprehensive layup SOP manual on the team wiki that integrated CAD dimensions, detailed ply schedules, and high-resolution photos of vacuum bagging sequences. I optimized things by breaking large 45° plies into smaller sections that were butt-joined, I removed the need for difficult relief cuts and reduced material waste.

Result: * **Efficiency:** We reduced our prepreg requirement from **200 feet to 60 feet** for the package—a 70% material savings.

- **Scalability:** Most importantly, we eliminated "tribal knowledge" errors, allowing junior members to reliably build flight-grade components with zero bridging defects across the entire season.

Questions:

1. Why is symmetry important in these parts?
 - a. Aerodynamic structures often rely on symmetry to maintain balanced load distribution and predictable aerodynamic behavior. If one component has slightly different fiber orientation or geometry than its mirror component, it can introduce uneven aerodynamic forces or structural responses. Standardizing the manufacturing process helps ensure both parts are built identically.
2. How do you ensure two composite parts are actually identical?
 - a. Standardized manufacturing
 - b. Consistent trimming and inspection

~~—Situation: Manufacturing cycles have a year gap between them and a lot of new members who don't have experience. This leads to inefficiency and reliance on "tribal knowledge".~~

~~—Task: The goal was to document part manufacturing processes this semester. Things like the exact blocks of foam for part, specific ply schedules, nested plies and things to keep in mind when cutting, laying up process and major things to focus on and such.~~

~~—Action: Developed and documented a comprehensive SOP manual on the team wiki integrating CAD dimensions of parts, step-by-step checklists, and high-resolution photos of the stock and tooling.~~

- I implemented digital ply nesting and switched from full-length cuts to part-specific ply kits, which reduced relief cuts and waste.
- Part cuts of plies instead of full-length cuts to reduce relief cuts + save material
- Reduced the total material requirement from 200 ft to just 60 ft for the front wing skin.
- Helped educate new hands on SOP, safety and so on.

Our 21-element aero package faced a significant gap between the design and manufacturing phases. With many new team members and a reliance on "tribal knowledge," we were seeing inconsistent part quality and massive material waste, particularly when cutting plies at a 45° orientation.

Task: Standardize the fabrication process for a 9-member team and optimize carbon fiber usage to stay within a tight budget.

Action: * Developed a comprehensive SOP manual on the team wiki, integrating CAD dimensions, step-by-step checklists, and high-resolution photos of the stock and tooling.

- Utilized **deepnest.io** to automate the nesting of parts from a single roll of carbon.
- Strategized the layup by breaking the 5 main wing sections into smaller subsections. Instead of cutting one massive 45° piece (which creates two large unusable triangles of waste), I designed the plies to be cut in smaller sections that were then butt-joined or overlapped during the layup.

Result: * Achieved a 65% carbon utilization rate.

- Reduced the total material requirement from **200 ft to just 60 ft** for the front wing skin.
- Established a replicable manufacturing workflow that eliminated errors caused by the manufacturing "gap" between seasons.

2.1.2 Structural Validation & FSI (FSAE)

- **Situation:** Legacy designs for our front wing relied on static "remote forces" to size structures. This failed to account for how the wing actually deflected under aerodynamic load, leading to parts that were either over-built (excess weight) or under-stiff (leading to unknown aerodynamic performance loss).
- **Task:** My goal was to implement a high-fidelity structural validation loop to find the optimal balance between component mass and aerodynamic efficiency.
- **Action:** I directed a Fluid-Structure Interaction (FSI) study by mapping CFD pressure fields directly onto our FEA mesh. This allowed me to iterate on the structures, ply schedules and material selection based on more realistic loads rather than approximations.
- **Result:** This workflow allowed us to reduce the wing weight by **0.6 lbs** while ensuring that peak deflection remained under **0.48"** at 95 mph. We successfully limited the CI loss to **0.04**. The parts were also designed to be more manufacturable.

2.1.3 DFM & Flat-Stock Optimization (FSAE)

- **Situation (The Production Risk):** During the design season for our latest aero package, there was a realization that manufacturing all the composite parts of the subteam would be difficult to manufacture in the timeline we had.
- **Task:** My goal was to redesign the internal structure to prioritize manufacturability, ensuring the team could reliably build parts and meet our strict production deadlines.
- **Action:** I transitioned the internal spars and ribs to be flat-stock composite components that could be manufactured as simple laminate plates and trimmed afterward. To ensure accurate assembly, I designed a **cross-lap joint system** with interlocking slots that acted as a self-aligning fixture. To compensate for the loss of geometric stiffness (from not using I/C beams), I optimized the laminate design by increasing the ply schedule in high-stress regions from three to five plies.
- **Result:** The redesigned structure significantly reduced layup complexity and build time, and the interlocking geometry eliminated previous alignment issues. This allowed the team to manufacture parts more reliably while still meeting all structural performance requirements.

Questions:

1. Why are flat laminates easier to manufacture?
 - a. Simpler ply placement, no tight radii and better control over fiber directions
2. What are the trade offs of flat stock structures?
 - a. Slightly less structurally efficient
 - b. Required cross contact for alignment compared to conforming to the wing surface.
3. How would you evaluate manufacturability earlier in the design process?
 - a. Consider tooling and layup paths
 - b. Simplify geometry

Situation: ~~In previous years, complex 3D internal geometries (ribs and spars) were the primary bottleneck in manufacturing, leading to missed deadlines and high scrap rates due to layup difficulty in tight corners.~~

Task: ~~Redesign the internal structural components to prioritize "buildability" and ensure the team could hit strict production milestones.~~

Action: * ~~Transitioned the internal structures from complex 3D shapes to flat-stock components.~~

- ~~Engineered a **Cross-Lap Joint** system (interlocking slots) that served as a self-aligning mechanical placer during the assembly of the internal skeleton.~~
- ~~Selected **Hysol EA 120** as the structural adhesive for the bond at the intersection points.~~

Result: * ~~To compensate for the loss of structural efficiency (moving away from I-beam/C-beam cross-sections), I **increased the ply schedule from 3 to 5 plies** in high-stress regions.~~

- ~~This trade-off significantly reduced total layup hours and eliminated the placement errors common in the previous 3D designs.~~

2.1.4 The Leading Edge Bridging RCA (Rear Wing)

- **Situation (The Quality Defect):** While manufacturing our rear wing mainplane in a split mold, we discovered structural bridging along the leading-edge radius. This was both a structural and aerodynamic problem for us because the leading edge didn't have clean geometry.
- **Task:** Upon looking at the situation and the vacuum bag, I realized what the problem was. My goal was to ensure that this didn't happen to any other part.
- **Action:** During debuggging, I identified that the vacuum bag was unusually taut across the leading edge. After inspecting the bag geometry relative to the mold, I realized the initial bag cut did not account for the vertical "draw" (depth) of the split mold, causing the bag to span the radius rather than conforming to it. I made note of this and revised the manufacturing SOP to add **intentional bag slack and pleats** at the leading edge to ensure the bag could reach the deepest part of the tool without tension.
- **Result:** We validated this process change during our front wing layup last weekend, which resulted in a high-quality part with zero bridging. This correction saved our production timeline because of the reduction in post processing and was a reminder that vacuum bag geometry is just as critical as the ply layup itself to ensure a solid part.

Questions:

4. How did you confirm the root cause was bag tension?
 - a. Location of bridge aligned with tension zone in bag
5. Why do pleats help prevent bridging?
 - a. Add local bag slack which is often required in concave geometry
6. What other factors could've caused bridging?
 - a. Tight radii
 - b. Poor ply draping
 - c. Incorrect ply sched for geom
7. How would this scale in a production environment?
 - a. Standardized bag templates
 - b. Marks on tooling of where pleats need to be added
 - c. Work instruction

~~**Situation:** After curing a rear wing in a split mold, we discovered structural bridging along the leading edge radius—a "no go" defect that creates resin rich zones and lacks fiber reinforcement.~~

~~**Task:** Perform a Root Cause Analysis (RCA) to identify the failure point and update the SOP to prevent recurrence across the remaining 20 aero elements.~~

Action: ~~* During the debugging process, I noticed the vacuum bag felt "short" and taut.~~

- ~~• Identified that the initial bag cut did not account for the "draw" (the vertical depth) of the bottom half of the split mold.~~
- ~~• Revised the SOP to mandate the use of **Pleats** (extra bag folds) at the leading edge to ensure the bag could reach the deepest part of the tool without tension.~~

Result: ~~* Eliminated bridging defects on all subsequent 20 elements of the aero package.~~

- ~~• Improved the team's overall understanding of vacuum bagging physics and the importance of bag slack in complex molds.~~

2.2 3D Printed Composite Characterization (Research)

- **Situation (The Technical Challenge):** I worked with a beta-model Continuous Fiber 3D (CF3D) printer to characterize a new, experimental resin system. Because the hardware was in a pre-production stage, we faced significant problems with tow lifting and fiber pullaway during toolpath turns.
- **Task:** My goal was to produce high-fidelity composite specimens that met strict ASTM standards (D3039 and D3518) to generate reliable material property data for future structural analysis.
- **Action:** I identified that the lifting was caused by insufficient green strength in the underlying layers. To resolve this consistent issue, I modified the manufacturing workflow to include pauses in the print cycle. By allowing the resin to reach a partial cure, before continuing, I ensured the substrate was tacky enough to anchor the new fiber deposition without lifting.
- **Result:** These process adjustments resolved the tow lifting issues and allowed us to fabricate specimens with consistent fiber placement and volume fraction. We successfully generated validated mechanical property data that now serves as the foundation for structural analysis models.
- Toray T1100- 12k and Ceremat

Questions:

1. Why is fiber volume fraction important in composites?
 - a. Determines stiffness and strength
 - b. Too little fiber is a weak laminate
2. What causes tow lifting in additive composite printing?
 - a. I think mostly because of a lack of tack when parts are partially cured
3. How does fiber alignment affect mechanical properties?
 - a. I mean that's the principle of composites. They're anisotropy
 - b. Strength is highest in the fiber direction.
4. How does this translate to AFP?
 - a. Both involve controlled fiber deposition
 - b. Both require process parameter optimization

Situation: ~~Working with a beta model Continuous Fiber 3D (CF3D) printer and a new Ceremat (dual cure acrylate) resin, we encountered frequent fabrication errors, including tow lifting and poor fiber cutting during the additive process.~~

Task: ~~Generate experimentally validated material property data following ASTM D3039 (Tensile) and ASTM D3518 (Shear) standards.~~

Action: ~~* Strictly followed ASTM standards to ensure the validity of the generated data.~~

- ~~Developed custom toolpaths in the CF3D software to control the **Ironing temperature** and deposition speed.~~
- ~~Focused on managing the resin flow rate at the nozzle to maintain a consistent **Fiber Volume Fraction** (V_f).~~

Result: ~~* Successfully resolved "tow lifting" issues, allowing for the high fidelity fabrication of structurally representative specimens.~~

- ~~Generated validated material data that serves as the foundation for future structural analysis using this additive technology.~~

2.3 Force Motors

Situation: High-RPM operational stability in a powertrain assembly was compromised due to net imbalance forces, which could lead to significant vibration and reduced bearing life.

Task: Reduce net imbalance forces by at least 80% to meet stability requirements for a high-performance rotating assembly.

Action: * Performed a **Vector Loop Analysis** to identify the magnitude and phase of existing imbalances.

- Iteratively calculated the placement and mass of counterweights using Python (NumPy), phasing them across the assembly to counteract both primary and secondary forces.
- Migrated legacy data into CATIA V5 to validate the physical placement of counterweights against the assembly's spatial constraints.

Result: * Achieved a **>90% reduction** in net imbalance forces.

- Significantly improved high-RPM stability and projected component lifespan, demonstrating an ability to apply analytical mechanics to physical hardware.

2.4 Gala Precision: Kaizen Conveyor Modification:

Situation

During my internship at Gala Precision, I spent time observing the shop floor even though most of my assigned work was office-based. I noticed that springs frequently **bounced out of a slanted funnel on the conveyor system during the stress-relieving process**, causing jams and forcing operators to stop the machine to clear them.

Task

My goal was to **reduce material spillage and eliminate the downtime operators were spending clearing these jams**.

Action

I conducted a **simple time study** and found that operators were losing about **20 minutes of production time per shift** clearing the jams.

I initiated a **Kaizen improvement project** and designed a **mechanical solution**: a custom lip along the edges of the funnel to guide the springs back into the conveyor path.

I modeled the modification in CAD and created a **Bill of Materials to justify the minimal cost of implementation**.

Result

By the end of my internship, the modification had been installed on **two out of the three machines**, and the spillage issue was eliminated.

This removed approximately **20 minutes of downtime per shift**, improving the overall efficiency of the production line.

Questions:

1. Why did you perform a time study first?
 - a. Justify the solution and quantify the impact
2. Why did you choose a passive mechanical solution?
 - a. Minimal cost and quick remedy
3. How would you evaluate the improvement?
 - a. Compare downtime before and after
 - b. Operator feedback

Situation: ~~While my primary internship tasks were office-based (procurement and suppliers), I frequently visited the shop floor and observed a recurring issue where springs were bouncing out of a slanted funnel on the conveyor line during the stress-relieving process.~~

Task: ~~Minimize material spillage and reduce the resulting daily downtime spent by operators clearing machine jams.~~

Action: * ~~Conducted a time study and determined that manual cleaning and jam resolution were costing 20 minutes of downtime per shift.~~

- ~~Initiated a **Kaizen** (continuous improvement) project to design a passive mechanical solution.~~
- ~~Drafted a CAD model for a **custom mechanical lip** for the conveyor edges and created a Bill of Materials (BOM) to justify the minor cost of implementation.~~

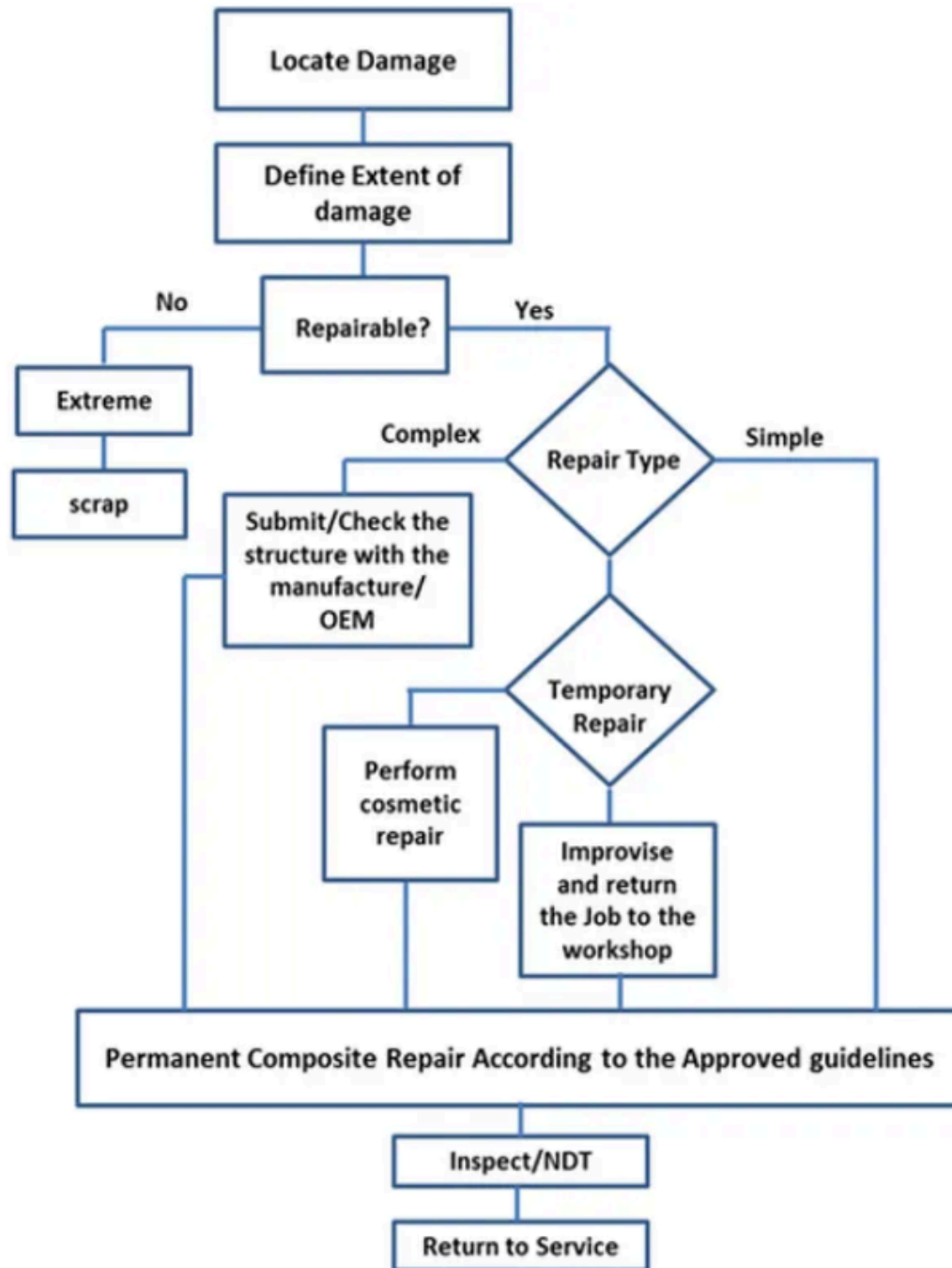
Result: * ~~Successfully installed the modification on two out of three machines by the end of my internship.~~

- ~~Eliminated the spillage issue and saved 20 minutes of production downtime per shift, demonstrating a proactive approach to shop floor efficiency.~~

5. Technical Deep Dive & Grilling Prep

Feature	Wet Layup	Resin Infusion	Prepreg	RTM
Process Overview	Dry fabric is placed on tool; liquid resin is applied manually with rollers/brushes.	Dry fabric is vacuum-bagged; vacuum draws liquid resin through feed lines from a reservoir.	Fabric comes "pre-impregnated" with semi-cured resin. Laid on tool and cured under heat/pressure.	Dry preform is placed in a two-sided, rigid "matched" mold. Resin is injected under high pressure.
Fiber Vol	30-40% (Low)	50-55% (Medium-High)	60-65% (High)	55-60% (High)
Void Content	High (>3%). High risk of trapped air.	Low (1-2%). Vacuum removes most air before resin entry.	Ultra-Low (<1%). Pressure "crushes" micro-voids.	Very Low. High pressure ensures full impregnation.
Structural Pros	Good for non-load	High stiffness to weight; good for	Gold Standard.	Excellent

	bearing fairings	large structural skin	Highest strength/stiffness repeatability.	dimensional accuracy on both sides of the part.
Mfg Pros	Lowest cost; no specialized equipment; fast for one-offs.	Scalable for large parts (wing skins); lower VOCs than wet layup.	High precision; "tacky" plies allow for complex vertical layups.	High production rate (cycle time); minimal "touch labor" once set up.
Mfg Cons	Highly operator-dependent; messy; inconsistent thickness.	Risk of "dry spots" if plumbing fails; high consumable waste.	High cost; requires freezer storage; long autoclave cycles.	Extremely expensive, heavy matched-metal tooling.
Required equip	Brushes, rollers, scale, PPE.	Vacuum pump, resin catch pot, spiral wrap, feed lines.	Autoclave or Oven, Freezer storage, Clean room.	Injection pump, Heated hydraulic press, Matched molds.
FSAE Example	Nothing	Nosecone and damper cover	Front wing and structural carbon parts	Nothing - not required
Joby Context		Possible for large, lower-stress housing components.	Primary Flight Structures (Fuselage, Wing Spars, Rotors).	High-rate production of brackets or small structural fittings.



3.1 Tooling & Materials

- **Frekote:** Semi-permanent release agent. Applied in thin coats; mirrors tool surface.
- **EA 120 / EA 9309:** Aerospace structural epoxies. High shear strength, requires abrasion/solvent prep.
- **Epibond:** Potting compound for honeycomb edge closeouts or insert reinforcement.
- **CTE (Coefficient of Thermal Expansion):** Matching tool CTE to part CTE (e.g., carbon tool for carbon part) prevents dimensional warping during the 250°F+ cure cycle.

- **Plain vs. Twill Weave:** Twill (2x2) is more "drapeable" for complex aero curves; Plain (1x1) is more stable but prone to wrinkling on curves.

3.2 David Pedraza's Likely Grilling Points

- **Traceability:** How do you track "Out-life" or batch numbers for flight hardware? (Answer: Freezer logs and batch stickers).
- **RCA (Root Cause Analysis):** "If you see a void in a radius, what is your first check?" (Answer: Check bag slack/pleats, then look for ply-overlap 'bridges').
- **Lean/Kaizen:** "Why was the conveyor lip a better fix than a sensor?" (Answer: Passive mechanical fixes are more reliable/durable in high-rate manufacturing).

6. Behavioral Interview Questions:

4.1 Conflict & Teamwork

- **Situation:** During a tight design timeline for our FSAE front wing, I was working with a junior member on the FEA setup. We needed to iterate quickly on the design to meet our manufacturing freeze.
- **Task:** Establish an efficient workflow for iterating on CAD geometry within our analysis software (Ansys).
- **Action:** I advocated for using Ansys's internal CAD tools to make small, iterative changes without the time-intensive process of re-importing and re-assigning boundary conditions. My teammate preferred using external CAD software for its better modeling tools, despite the longer import cycle. After sensing some hostility in our text communications due to the high-stress environment, I scheduled a full-team meeting to realign on our shared goal. I then had a one-on-one conversation with the individual to "clear the air," acknowledging the stress of the timeline and validating his opinion while explaining the data-driven need for iteration speed.
- **Result:** We moved forward on the same page. By resolving the interpersonal friction early and explaining the "why" behind the workflow, we maintained our timeline and successfully hit our design freeze.

4.2 Initiative & Problem Solving

- **Situation:** During my internship at Gala Precision, my primary tasks were office-based (procurement and suppliers). However, I frequently visited the shop floor and noticed an issue with the stress-relieving machines.
- **Task:** Improve the flow of springs coming off the conveyor belt to reduce spillage and machine jams.

- **Action:** I observed that springs were bouncing out of a slanted funnel and onto the machinery, causing 20 minutes of daily downtime. Although it wasn't my assigned task, I proposed a "Kaizen" fix to my boss. He asked for a CAD model and a Bill of Materials (BOM). I drafted the mechanical "lip" design and identified the necessary parts for installation.
- **Result:** By the end of my internship, I had helped install the modification on two of the three machines. This passive mechanical fix eliminated the spillage and directly improved the shop's daily production efficiency.

4.3 Failure & Accountability

- **Situation:** Early in the season, we were laying up a rear wing in a complex split mold. To help train new members, I had them assist with pressing the carbon into the leading-edge radius.
- **Task:** Ensure the carbon was perfectly consolidated in the tightest crevice of the tool (the leading edge).
- **Action:** Upon demolding, we discovered bridging at the leading edge. My initial guess was that the hand-pressure from the new members wasn't enough. However, I conducted a Root Cause Analysis (RCA) while debuging and realized the bag itself was taut. The bag cut hadn't accounted for the "draw" of the mold. Instead of blaming "technician error," I identified it as a process flaw.
- **Result:** I updated our SOP to mandate **Pleats** on the vacuum bag at the leading edge. This experience taught me that in manufacturing, knowing "what not to do" is just as valuable as knowing the right steps. No subsequent parts had bridging defects.

4.4 Why did you do _?

7. **Situation:**
8. **Task:**
9. **Action:**
10. **Result:**

11. Joby Aviation Company Research

- **Mission:** Save a billion people an hour a day through quiet, electric flight.
- **Vertical Integration:** Joby builds everything in-house (Marina, CA) to control quality and certification timelines.
- **Manufacturing Shift:** Moving from "Prototype" to "Certification and High-Rate Production."
- **David Pedraza Connection:** Focus on precision (his watchmaking hobby) and the transition from SpaceX-style rapid prototyping to FAA-certified production standards.

12. Technical Case Study: “Explain how you would design a mold for a laminated part”

This is the specific question a previous Joby candidate received. Don't just list steps; explain the engineering logic.

Step 1: Material Selection (CTE Match)

- **Question:** Why choose one tool material over another?
- **Answer:** "I first consider the **CTE (Coefficient of Thermal Expansion)**. If I'm curing a carbon fiber part at 250°F, an Aluminum tool will expand significantly more than the part. Upon cooling, the tool shrinks more than the part, inducing internal stresses or warping. For aerospace tolerances, I would design a **Carbon Fiber Tool** (Composite-to-Composite) to ensure the CTE is matched, or use **Invar** for extremely high-volume production."

Step 2: Master Pattern & Flanges

- **Logic:** You need a "Plug" to pull the "Mold."
- **Detail:** "I would CNC machine a master pattern from a high-density tooling board (like RenShape). I'd ensure the flanges are wide enough (at least 2-3 inches) to provide a stable surface for the **Tacky Tape** and vacuum seal, and to prevent the bag from 'pulling' on the part edges."

Step 3: Surface Finish & Release

- **Logic:** The part is only as smooth as the tool.
- **Detail:** "The pattern would be sealed with a high-temp sealer (like Duratec) and sanded to 2000 grit. I'd apply a semi-permanent release agent like **Frekote**. I'd choose Frekote over wax because it's a 'chemical' release that doesn't build up in corners, preserving the aerodynamic profile of the airfoil."

Step 4: Air-Pathing & Vacuum Integrity

- **Logic:** You can't have a vacuum if air is trapped.
- **Detail:** "I'd design the mold with integrated vacuum ports or ensures 'breather' paths lead to every deep pocket. For a complex Joby part, I'd consider a **Split Mold** (Clamshell) with indexed pins to ensure the two halves align perfectly for a seamless leading edge."

13. Technical Intuition: Comparison Guide

Aluminum vs. Carbon Fiber Stress-Strain Curves (SpaceX Question)

- **Aluminum:** Ductile. It has a linear elastic region, a yield point, and then a long plastic deformation region before failure. It "bends before it breaks."
- **Carbon Fiber:** Brittle and Anisotropic. It is linear elastic nearly all the way to failure. It does not "yield"; it fails catastrophically once the fibers snap or the matrix cracks.
- **The "Intuition":** Aluminum is predictable and isotropic (same properties in all directions). Carbon is a "designed material"—you can make it 10x stronger in one direction than the other, which is why it's perfect for Joby's weight-sensitive rotors.

The "Lathe Cube" Logic (Skyroot Question)

- **Question:** How to cut a 50x50 cube from a cylinder on a Lathe?
- **The Trick:** Lathes rotate parts. You can't easily cut a "cube" purely on a lathe without a 4-jaw chuck and offset turning (turning one face at a time).
- **The "Manufacturing Mindset":** "While I could use a 4-jaw chuck to offset and turn four flat faces, a more efficient manufacturing approach would be to face/center-drill the cylinder on the lathe, then move it to a **3-axis Mill** using a vice to cut the square profile." (This shows you know when to switch machines).

14. High-level resources for Aerospace Manufacturing

To move beyond the hobbyist level, check these out:

1. **SAMPE (Society for the Advancement of Material and Process Engineering):** Their "Technical Papers" are the gold standard for what SpaceX and Joby actually do.
2. **CompositesWorld (Website/Magazine):** Look for their "High-Rate Production" and "Automated Fiber Placement (AFP)" articles.
3. **YouTube - "The Efficient Engineer":** Watch his videos on **Stress Tensors** and **Composite Mechanics**.
4. **Book:** *Manufacturing Processes for Advanced Composites* by F.C. Campbell. This is the "Bible" for aerospace manufacturing engineers.
5. **NASA Technical Reports Server (NTRS):** Search for "Composite Delamination" or "Autoclave Processing." It's free and is the exact data Joby engineers look at.